

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-09-13

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- *CSP Partial cross section data*
- *CST Temperature dependent cross section*
- *ETA Average neutron yield per nonelastic event*
- *NU Fission-neutron yield, nu-bar*
- *TKE Total kinetic energy*
- *SIG Cross section data given with limit as DATA and DATA-MIN*
- *SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)*
- *PFNS Prompt-fission neutron energy spectra*

-- Currently implemented search: by full-Reaction-code, DatasetID, Author1, product

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

- *CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state*
- *NU vs ENDF MT:452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event*
- *ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$*
- *PFNS vs ENDF - Prompt-fission neutron energy spectra (ratio to Maxwellian: $T=1.32\text{MeV}$)*

Code reac1.py help:

```

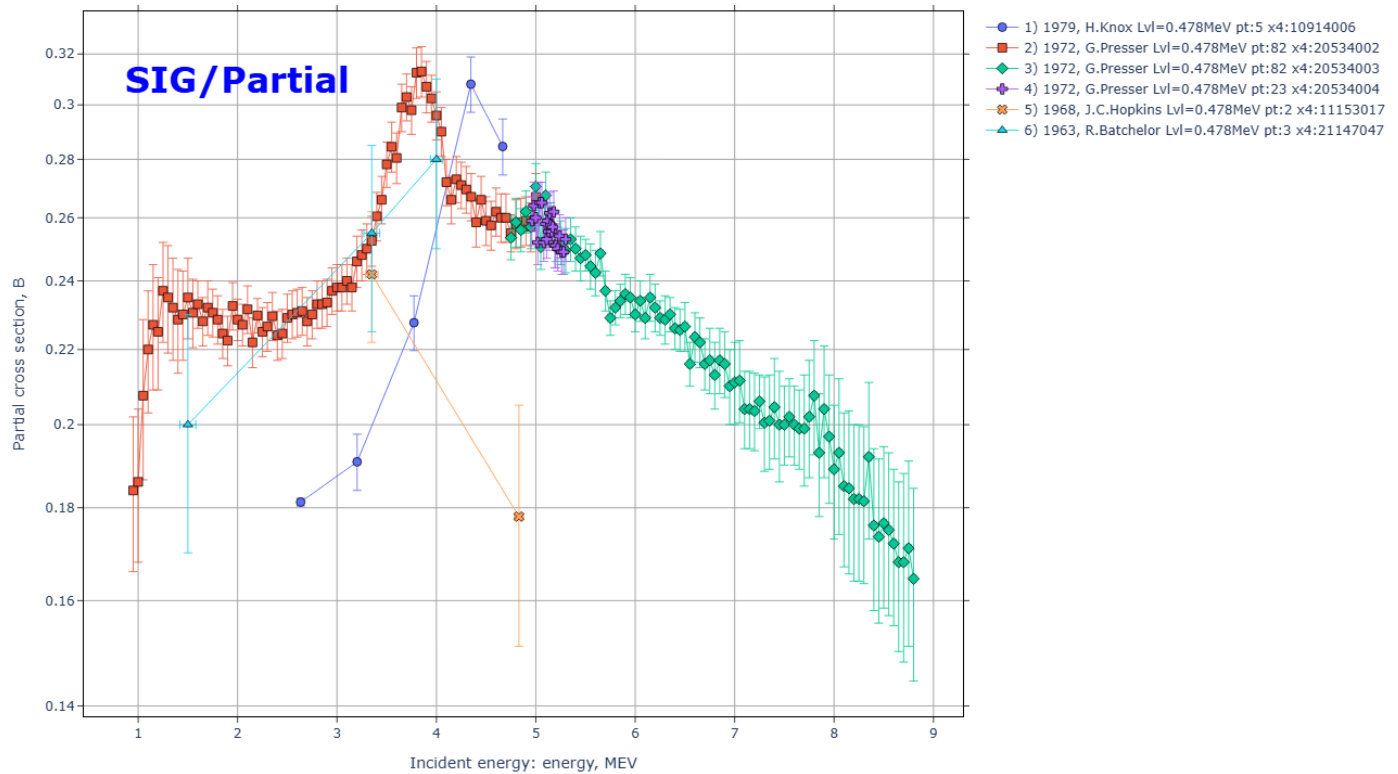
Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
  * computational data are presented as function y=f(x1,x2,..)
  where f is measured quantity, xi are independent variables:
      EN - incident energy,          E2 - energy of outgoing particle,
      ANG - angle of outgoing particle,  EVL - energy level, etc.
Algorithm:
  * user defines EXFOR Reaction-code for data search, x-variable by number
  * selects x-variable by number (default: x1)
  * other variables will be used as parameter for grouping of data points
Usage:
  $ python [{flag}] x4update.py [{option|reaction}]
  * flag: see all Python flags: $ python --help
        -B      don't write .pyc files on import
  * option:
    -help      print this help-text and exit (also --help)
    -o:<file>  output file name
    -x:x?      set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
    -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
    -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
    -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
    -nogrp     avoid grouping datasets by reaction-codes on the plot
    -fx:<num>  set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
    -fy:<num>  set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
  plotting:
    -xlog      logarithmic scale of X-Axis
    -ylog      logarithmic scale of Y-Axis
    -lines     connect points on the plot by lines
    -sym       draw border of a symbol of a data point on the plot
    -annot:x,y,txt annotation on top of the plot;
                x and y should be given in units of the plot;
                text can include html tags: <b>, <sup>
  * reaction:
    <reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"

```

part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

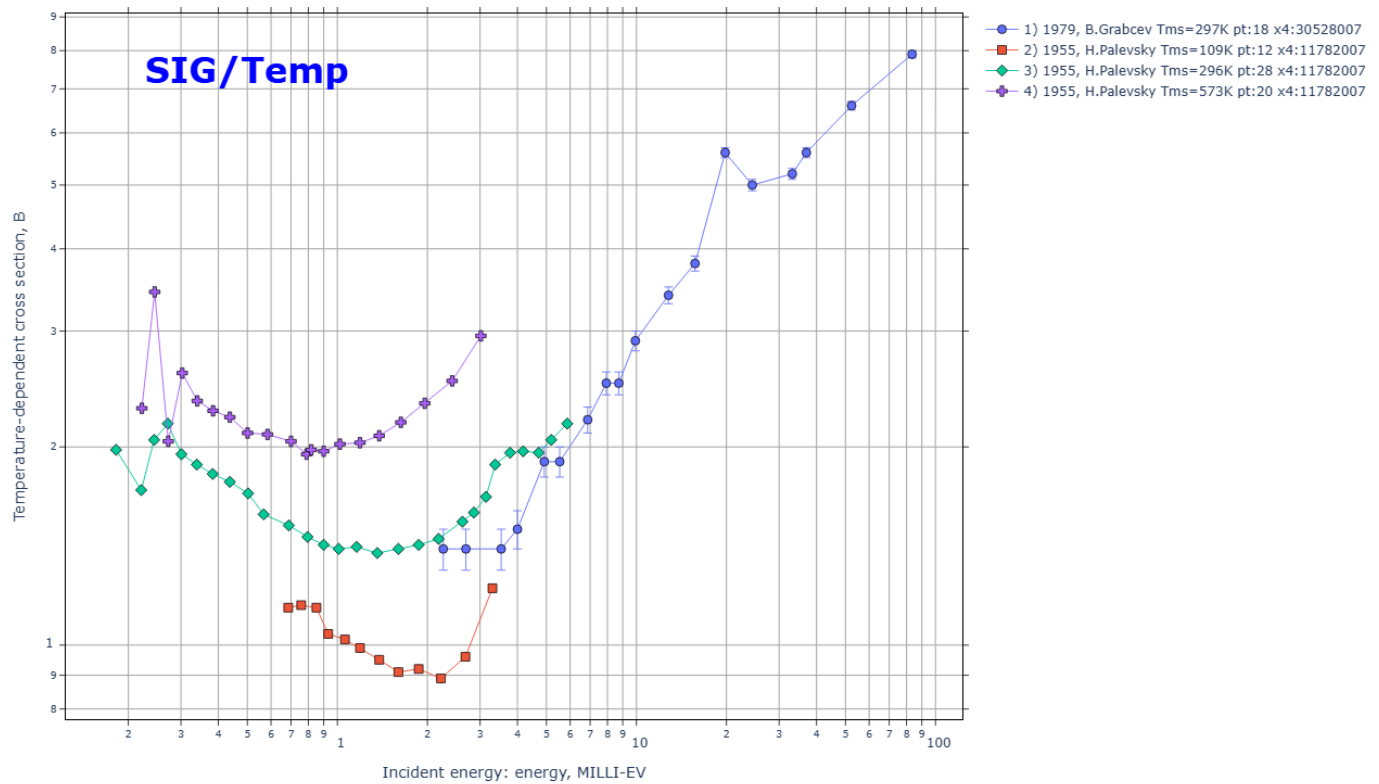
1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data

Reaction: 3-LI-7(N,INL)3-LI-7,PAR,SIG Quantity: CSP:y=Data(En,Lvl) Datasets:6 Points:197
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:45



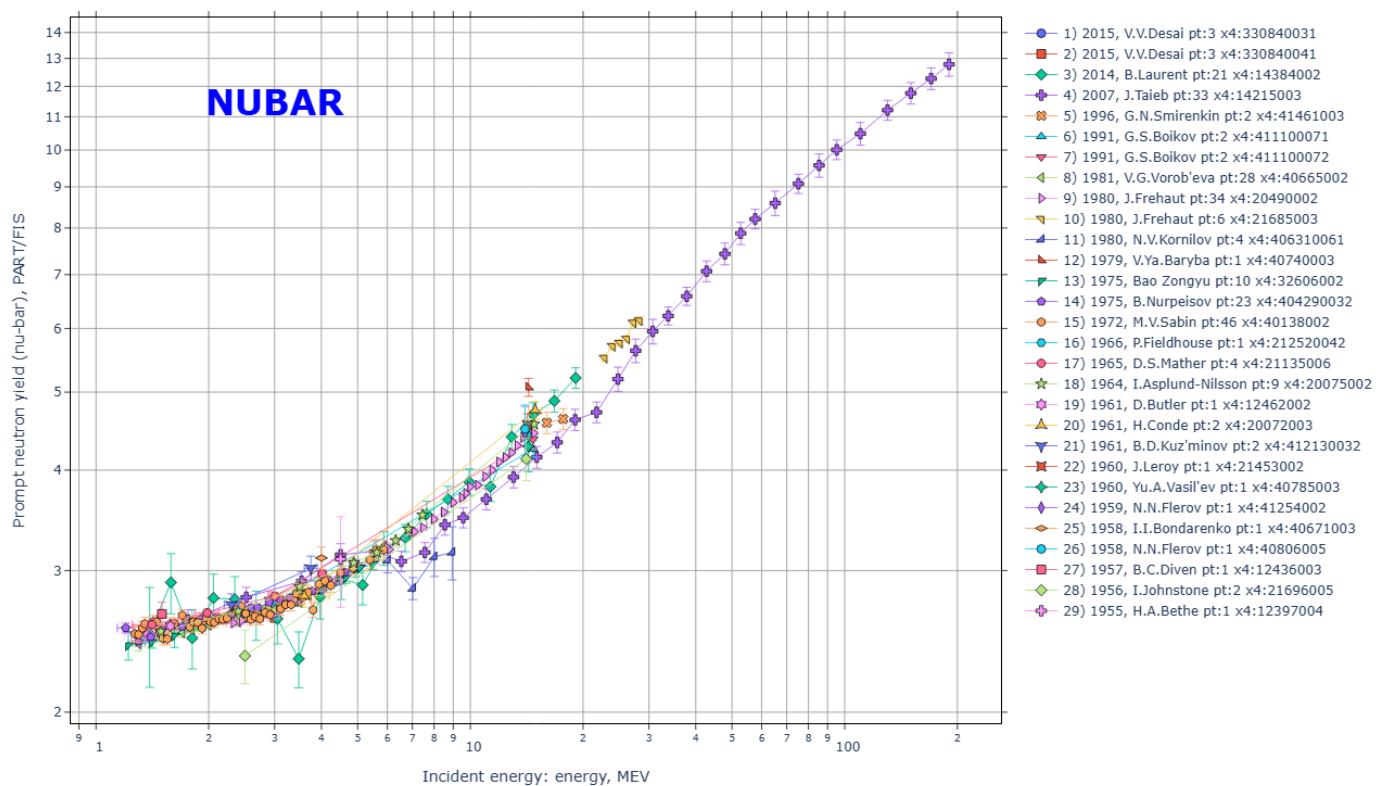
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction: 82-PB-0(N,TOT),,SIG/TMP Quantity: CST:y=Data(En,Tms) Datasets: 4 Points: 78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:33



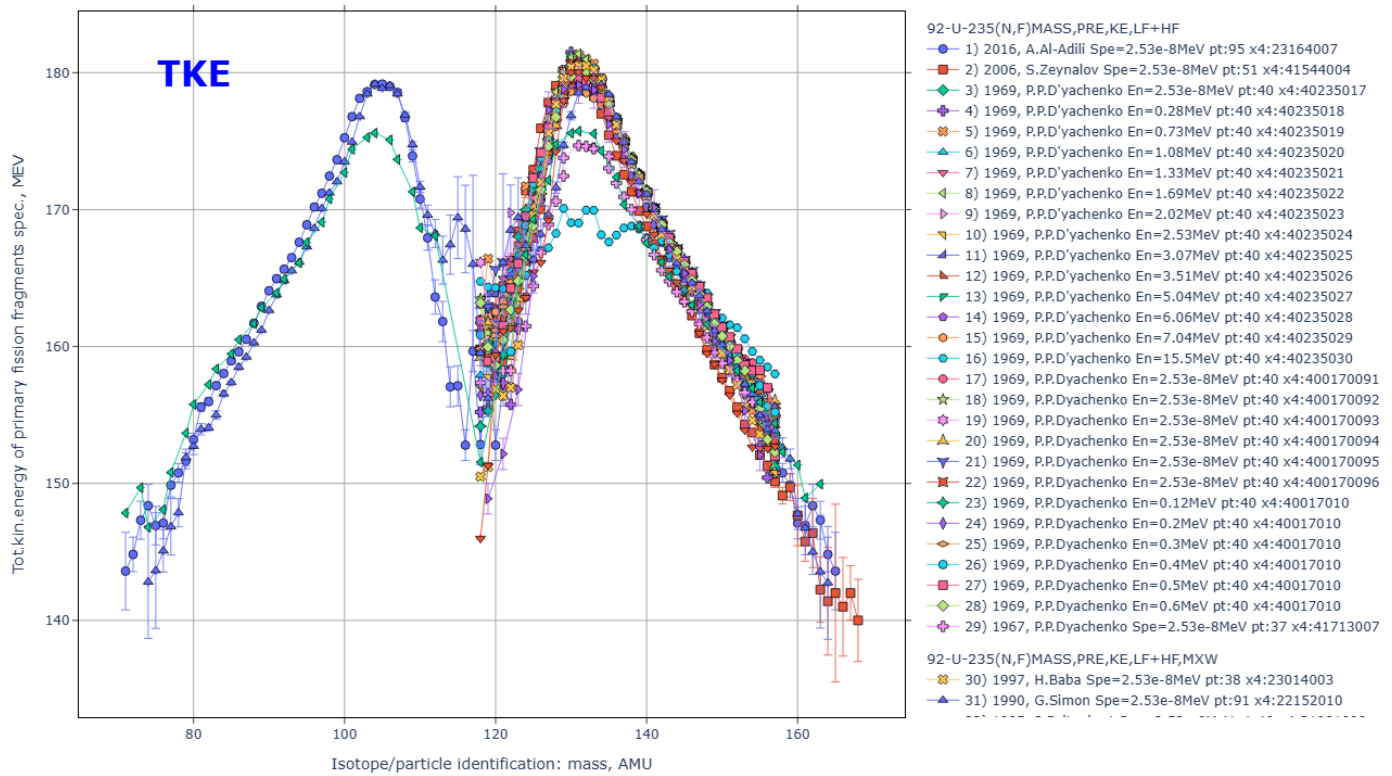
3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:36



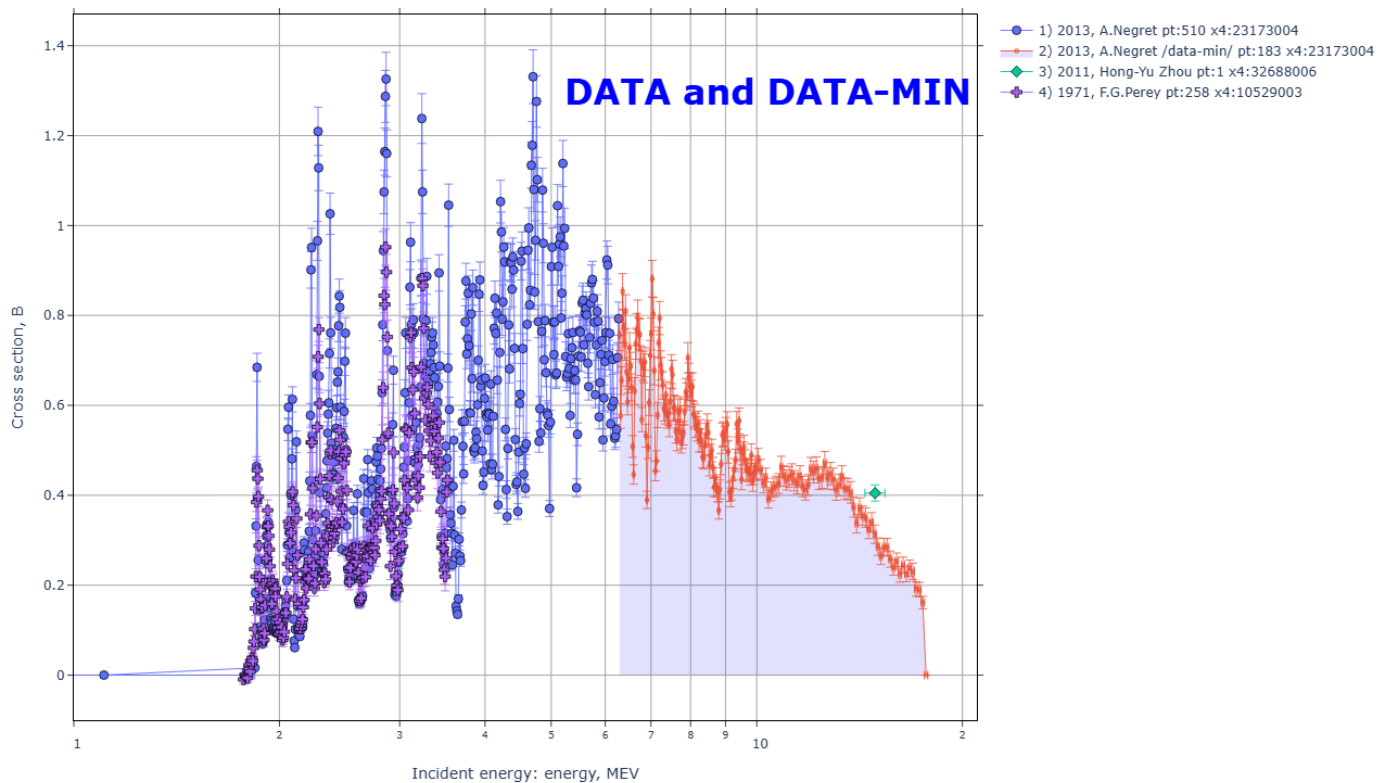
4) part1-8-reac/out00/tke1.html.png TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PRES,KE,LF+HF Quantity: E:y=Data(Spe,Zam) Datasets:34 Points:1467
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:39



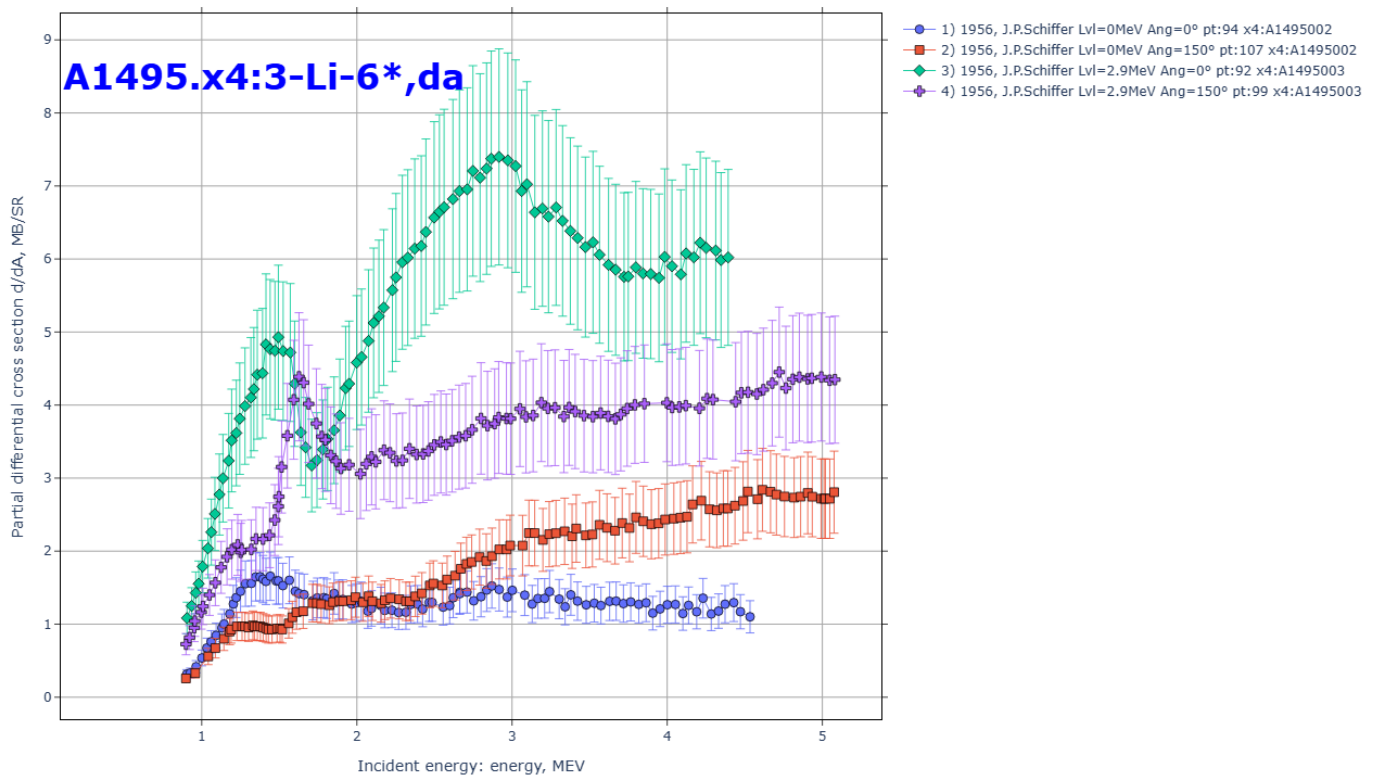
5) part1-8-reac/out00/si28nn.html.png SIG Cross section data given with limit as DATA and DATA-MIN

Reaction: 14-SI-28(N,INL)14-SI-28,,SIG Quantity: CS:y=Data(En) Datasets:4 Points:952
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:48



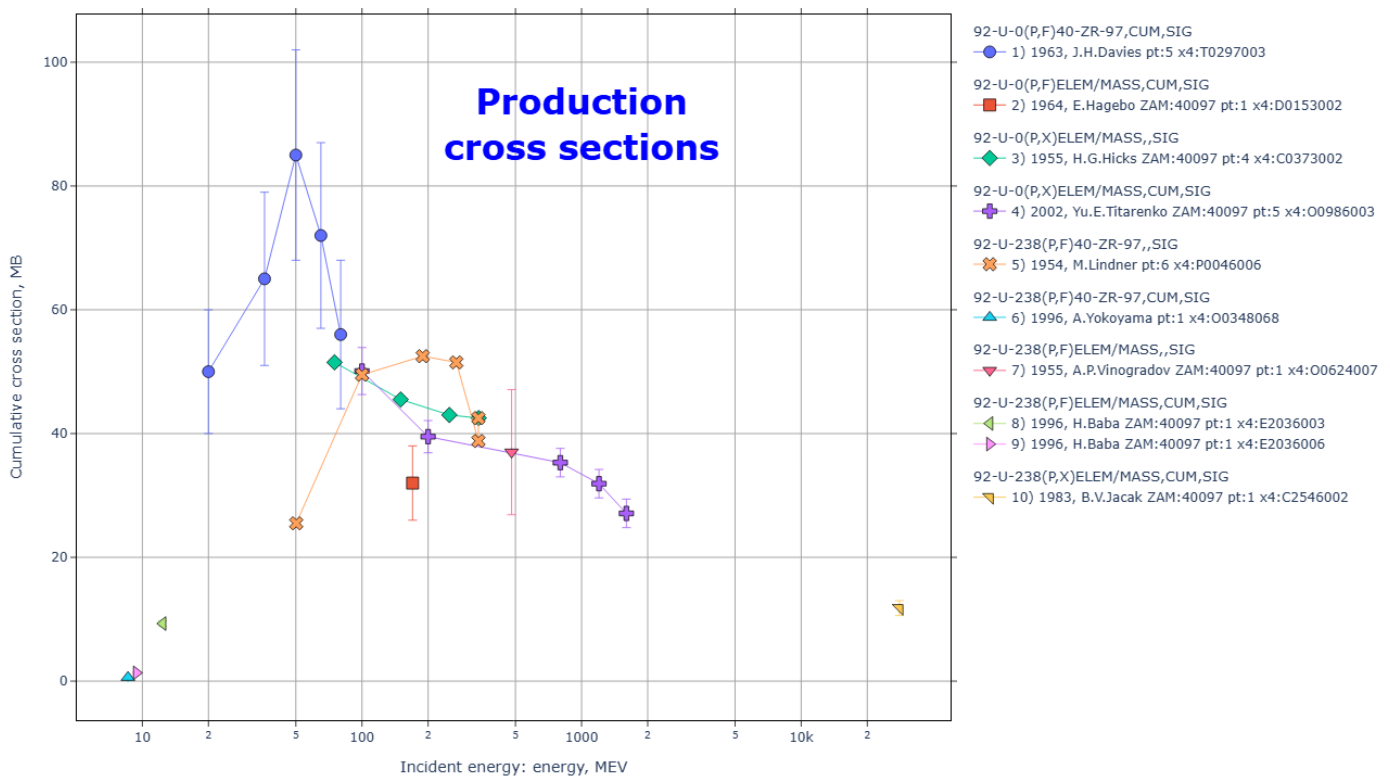
6) part1-8-reac/out00/a1495-li6.html.png Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

Reaction: 3-LI-6(HE3,P)4-BE-8,PAR,DA Quantity: DAP:y=Data(En,Lvl,Ang) Datasets: 4 Points: 392
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:50



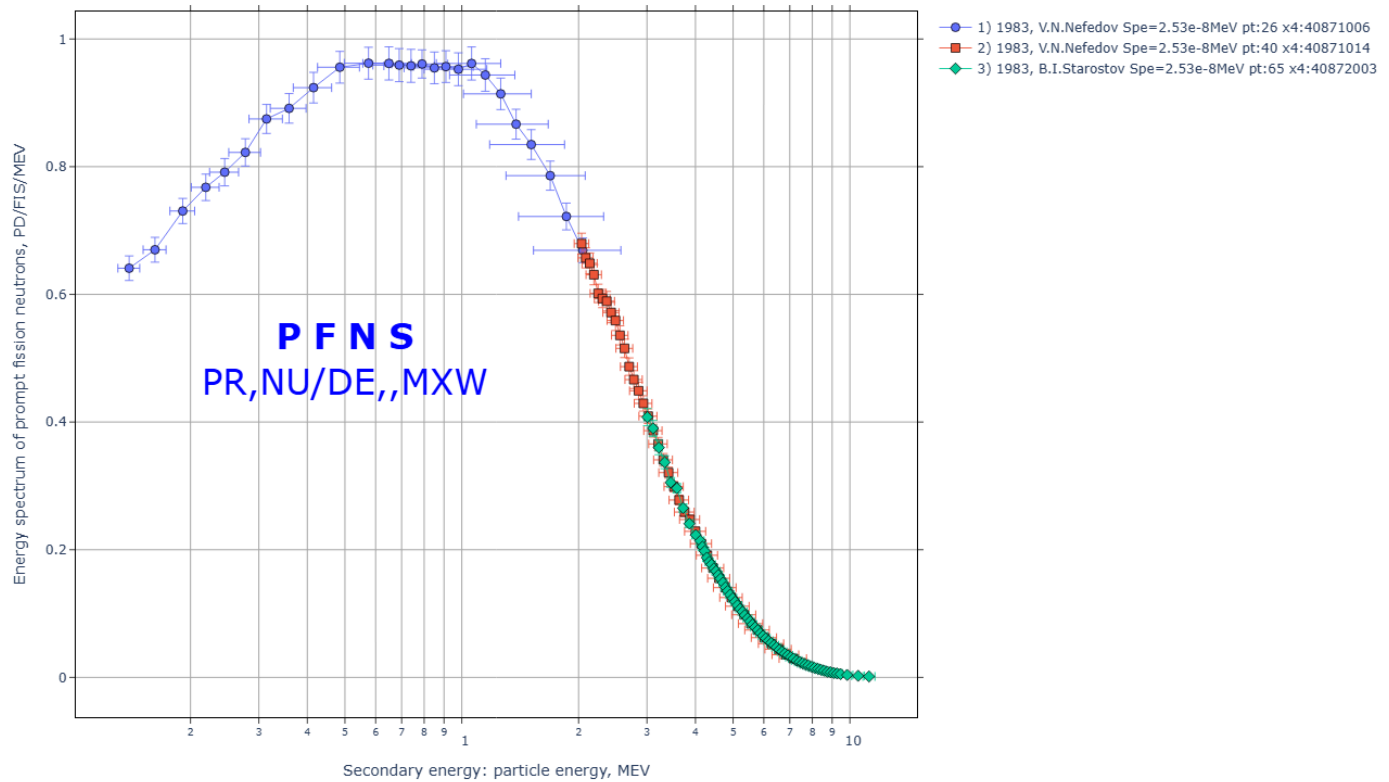
7) part1-8-reac/out00/sig1prod.html.png Production cross section coded as SF4 and ELEM/MASS

Reaction: 92-U-0(P,F)40-ZR-97,CUM,SIG Quantity: CS:y=Data(En) Datasets: 10 Points: 26
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:59



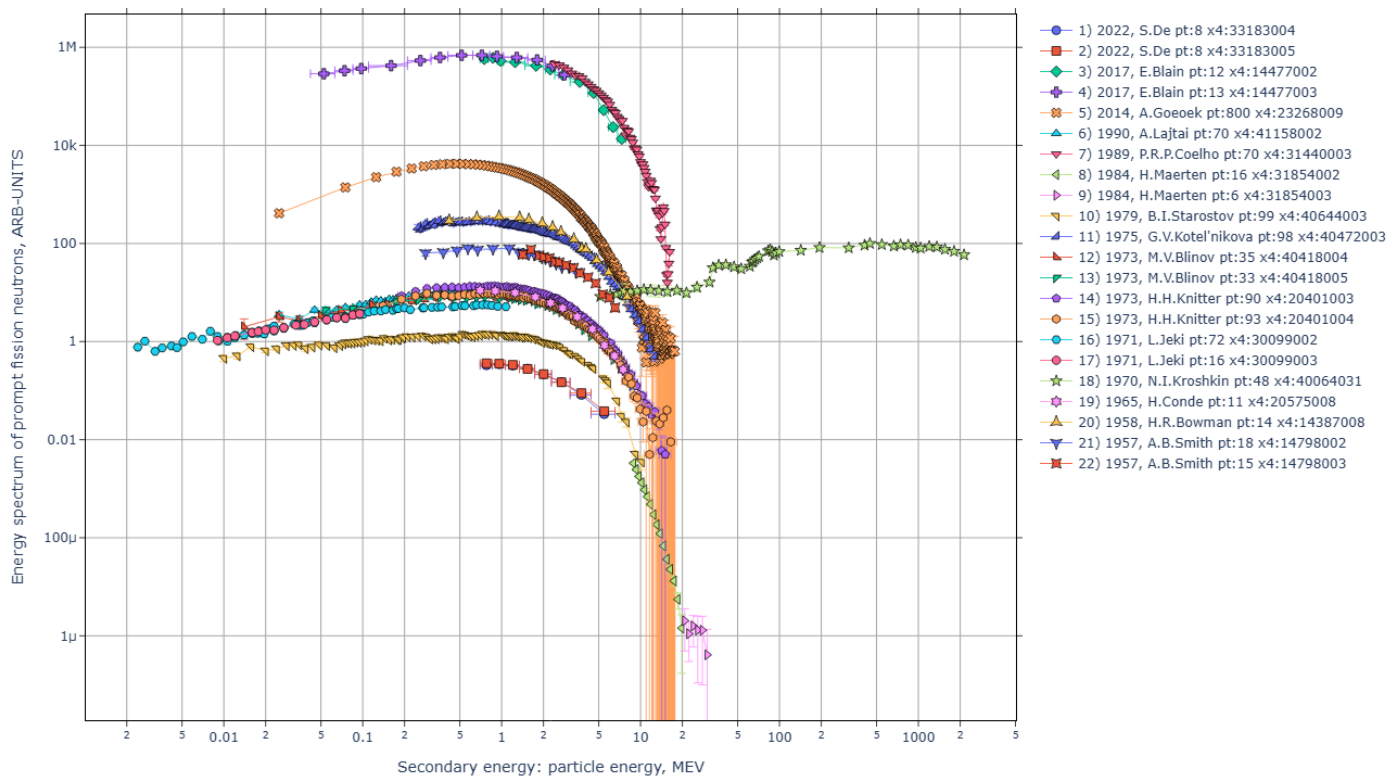
8) part1-8-reac/out00/pfns1.html.png PFNS Prompt-fission neutron energy spectra

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:43:01



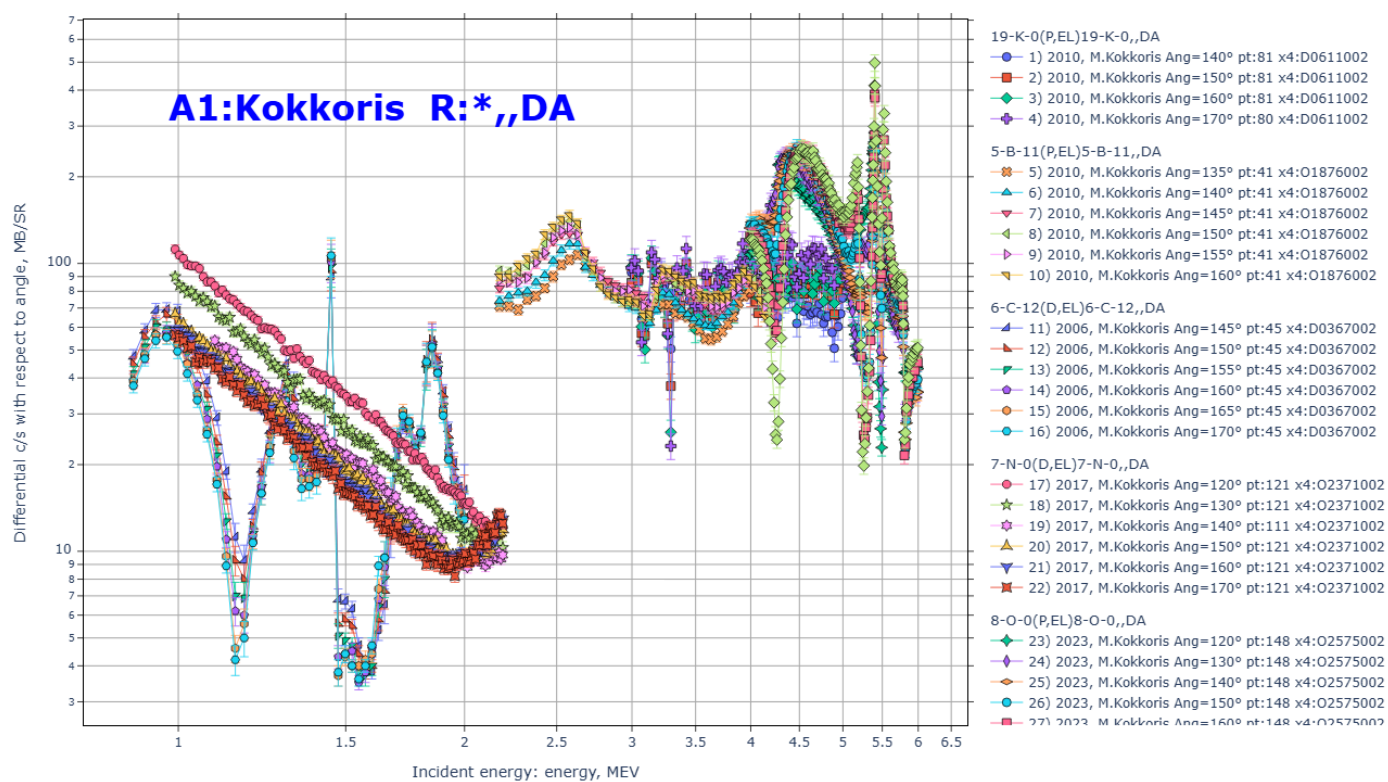
9) part1-8-reac/out00/pfns4.html.png Energy spectrum of prompt fission neutrons: relative data

Reaction: 98-CF-252(0,F),PR,NU/DE,,REL Quantity: MFQ:y=Data(E2) Datasets: 22 Points: 1645
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:43:09



10) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,DA"

Reaction: 19-K-0(P,EL)19-K-0,,DA Quantity: DA:y=Data(En,Ang) Datasets: 28 Points: 2443
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running: 2026-09-13 17:42:56

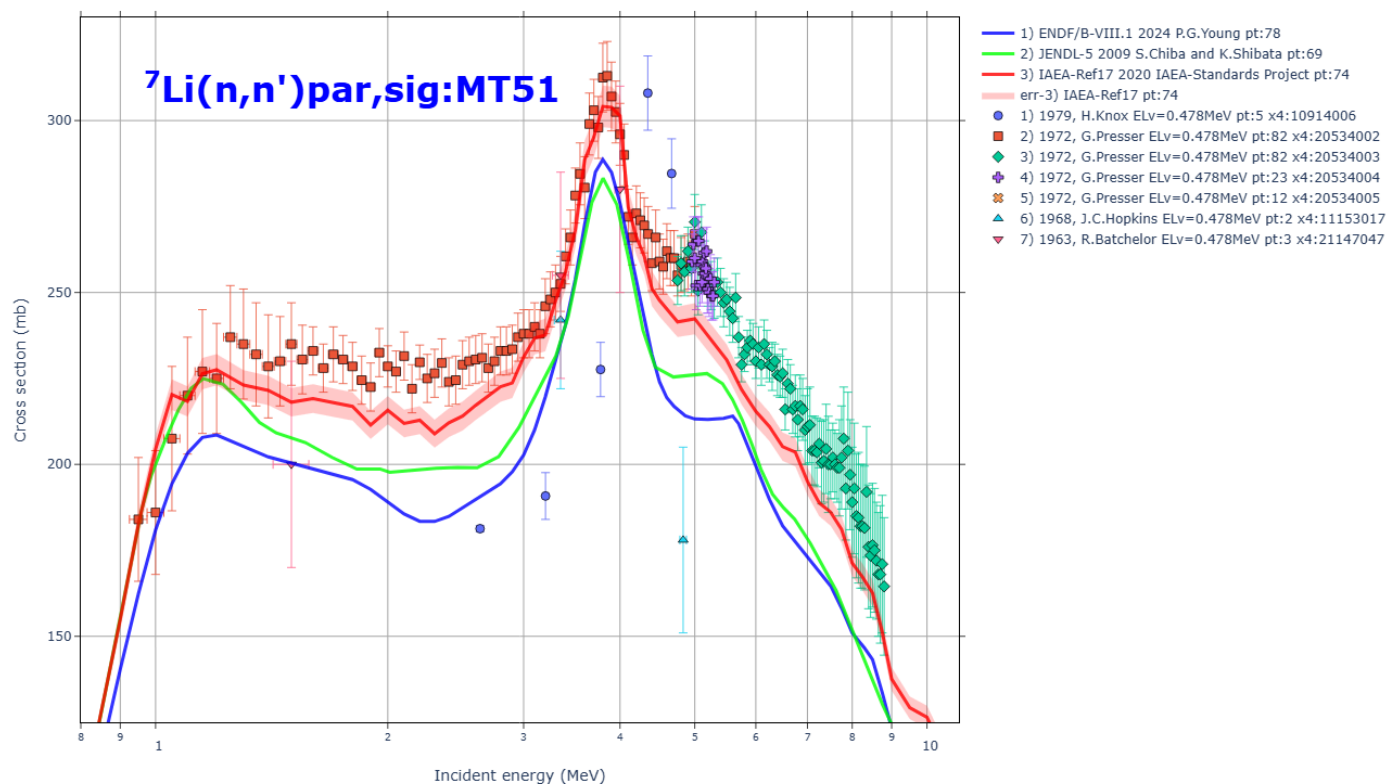


part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

11) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

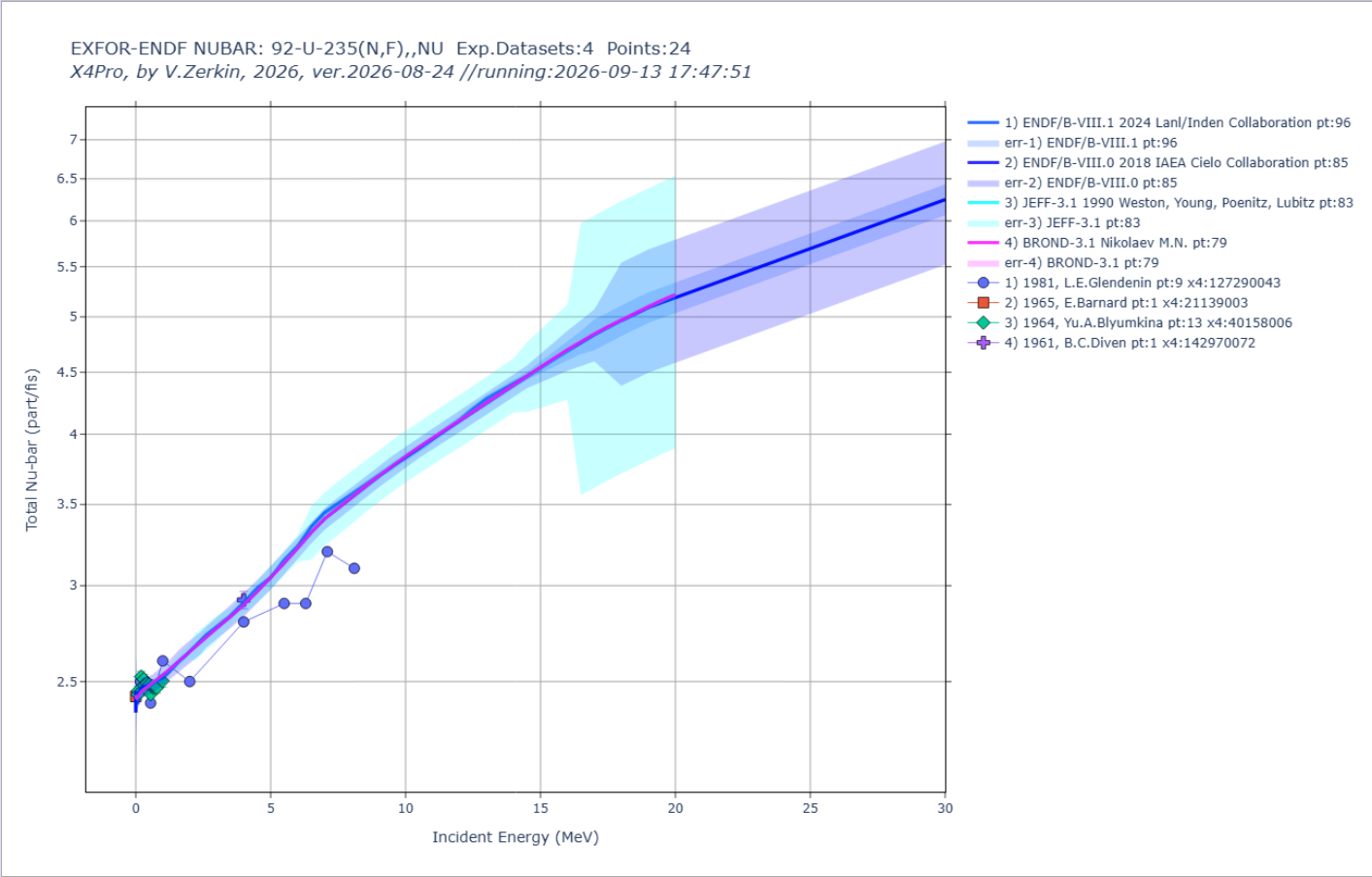
EXFOR/ENDF cross sections SIG(E): Li-7(n,nl)par,sig EXFOR-datasets:7

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2026, ver.2026-08-10 //run:2026-09-13 17:47:04



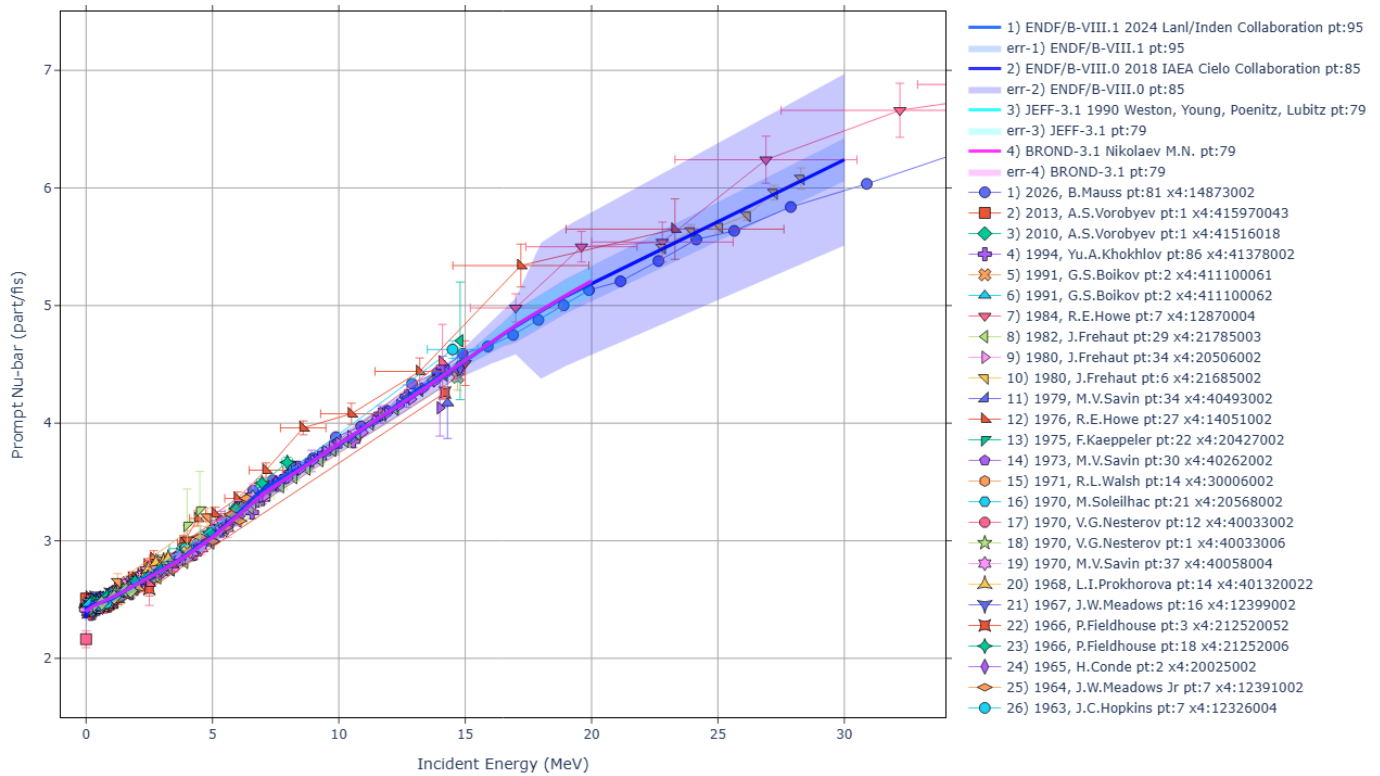
12) part2-9-nubar/out00/U235nf_-nu.html.png

Total neutron yield (nu-bar)



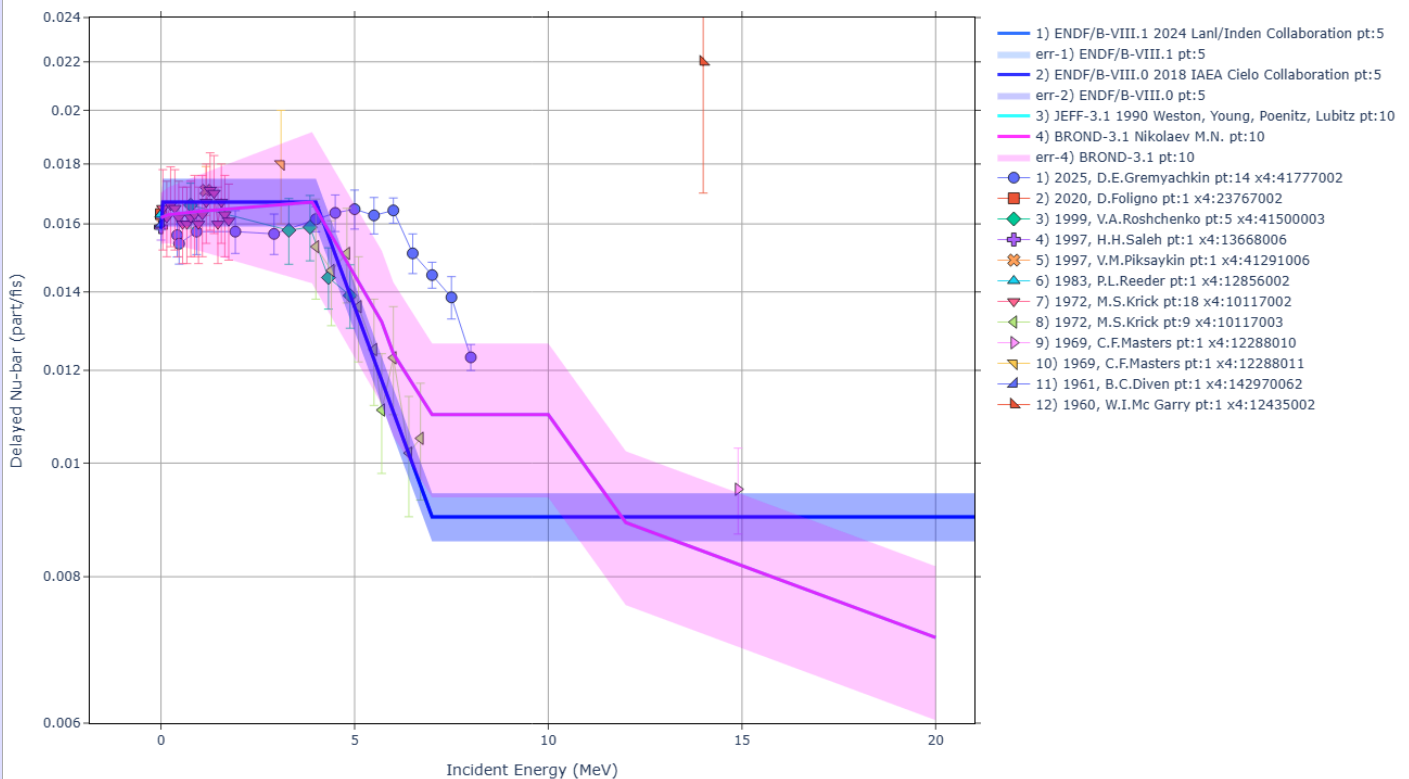
13) [part2-9-nubar/out00/U235nf_pr-nu.html.png](#) Prompt neutron yield (nu-bar)

EXFOR-ENDF NUBAR: 92-U-235(N,F),PR,NU Exp.Datasets:38 Points:535
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-13 17:47:17

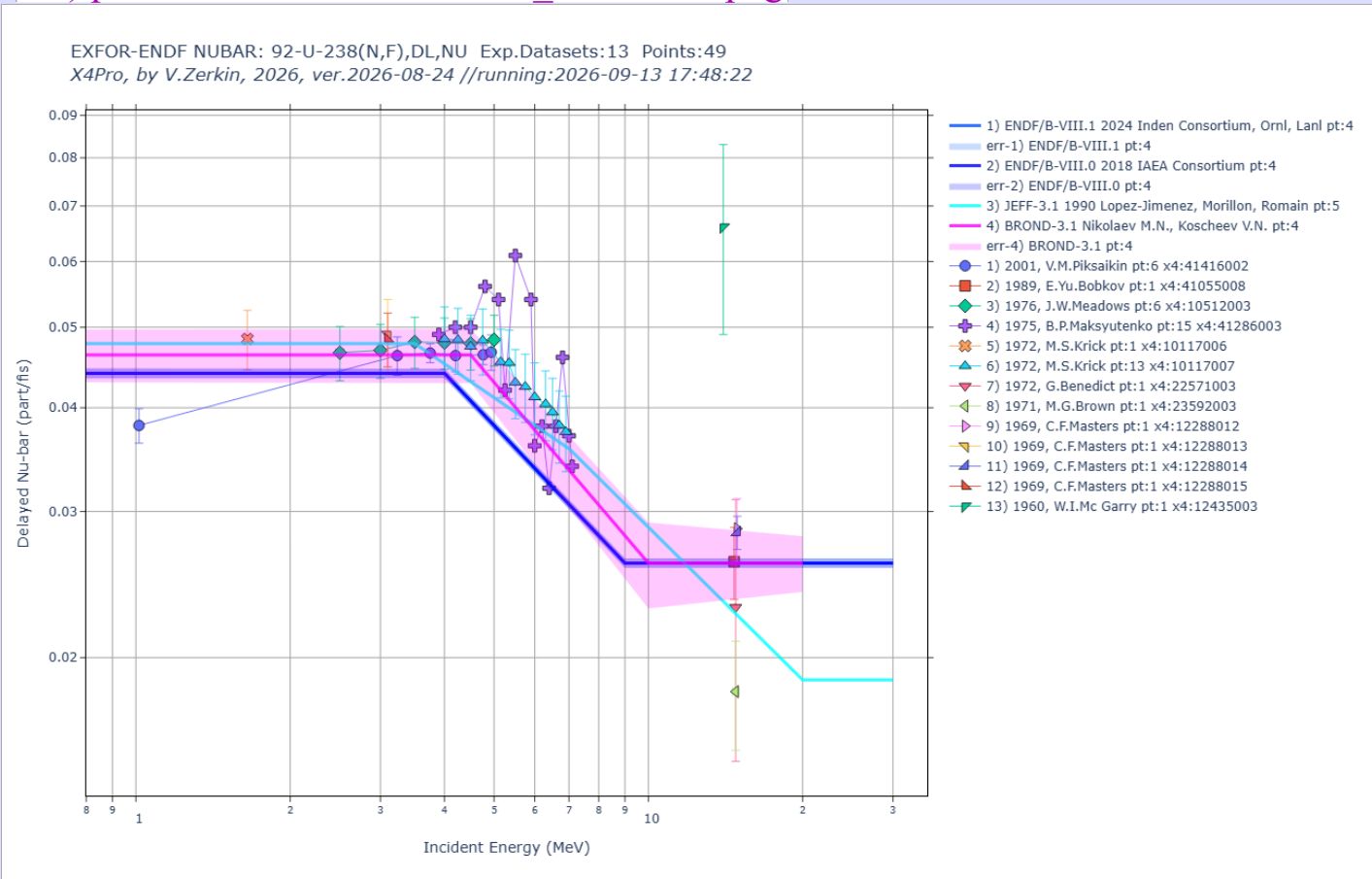


14) [part2-9-nubar/out00/U235nf_dl-nu.html.png](#) Delayed neutron yield

EXFOR-ENDF NUBAR: 92-U-235(N,F),DL,NU Exp.Datasets:12 Points:54
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-13 17:47:35



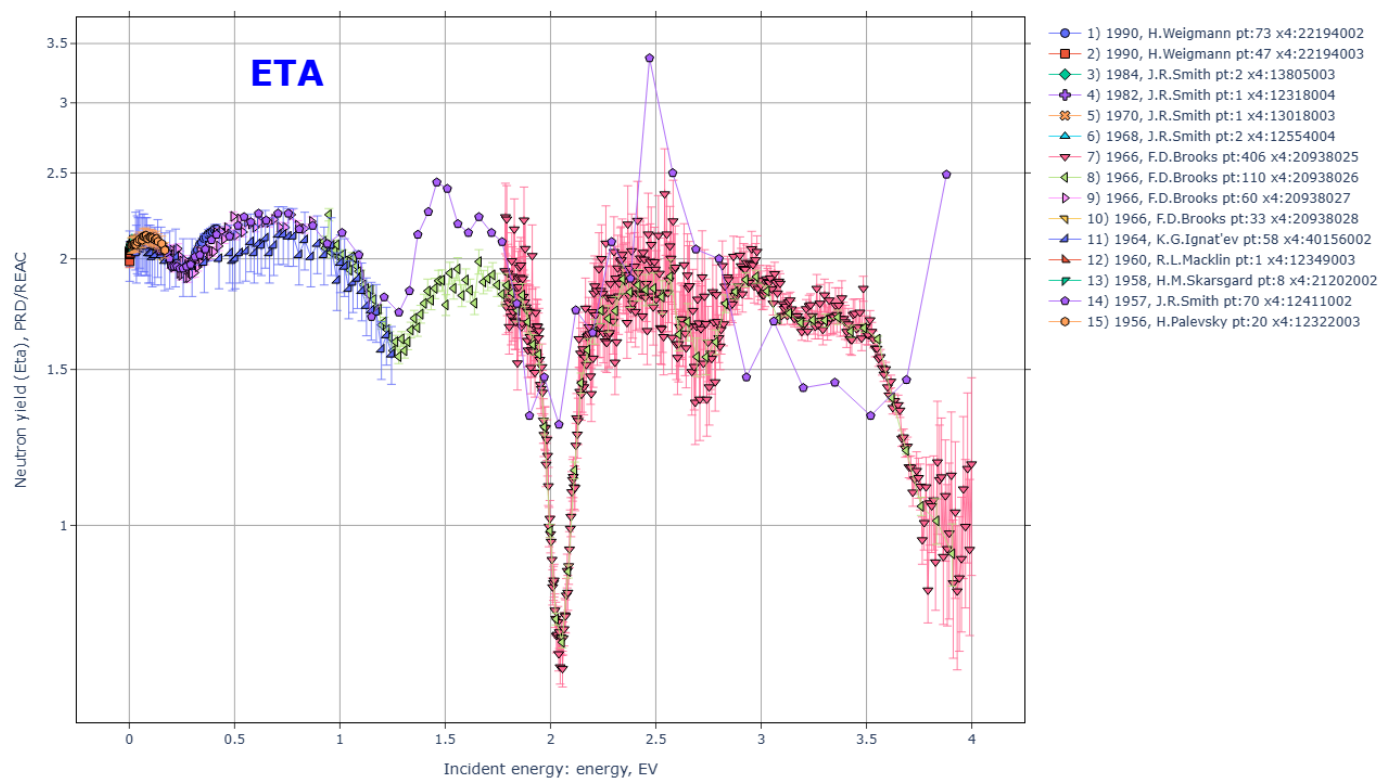
15) part2-9-nubar/out00/U238nf_dl-nu.html.png



part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

16) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

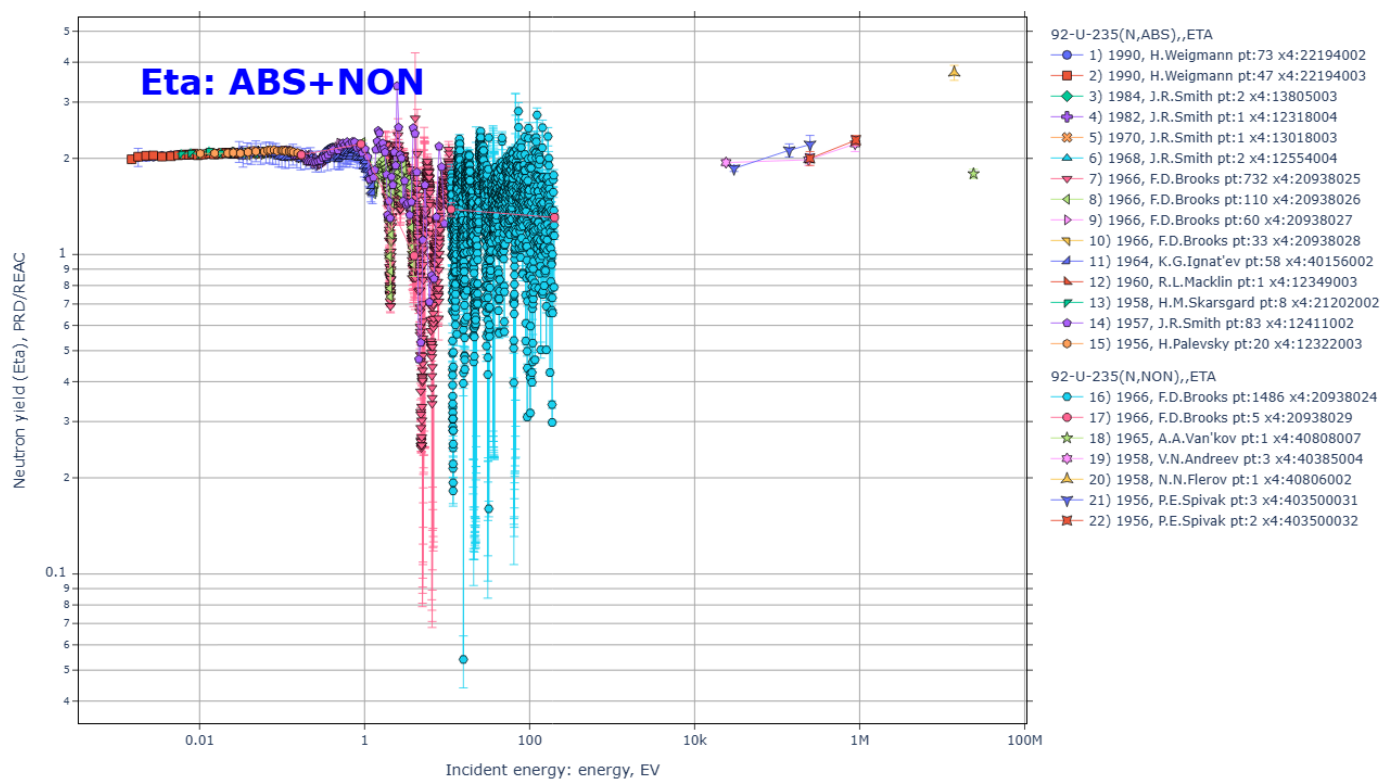
Reaction: 92-U-235(N,ABS),,ETA Quantity: NU:y=Data(En) Datasets: 15 Points: 892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-08-28 13:26:55



17) part1-8-reac/out00/eta2.html.png

ETA: (N,ABS)+(N,NON)

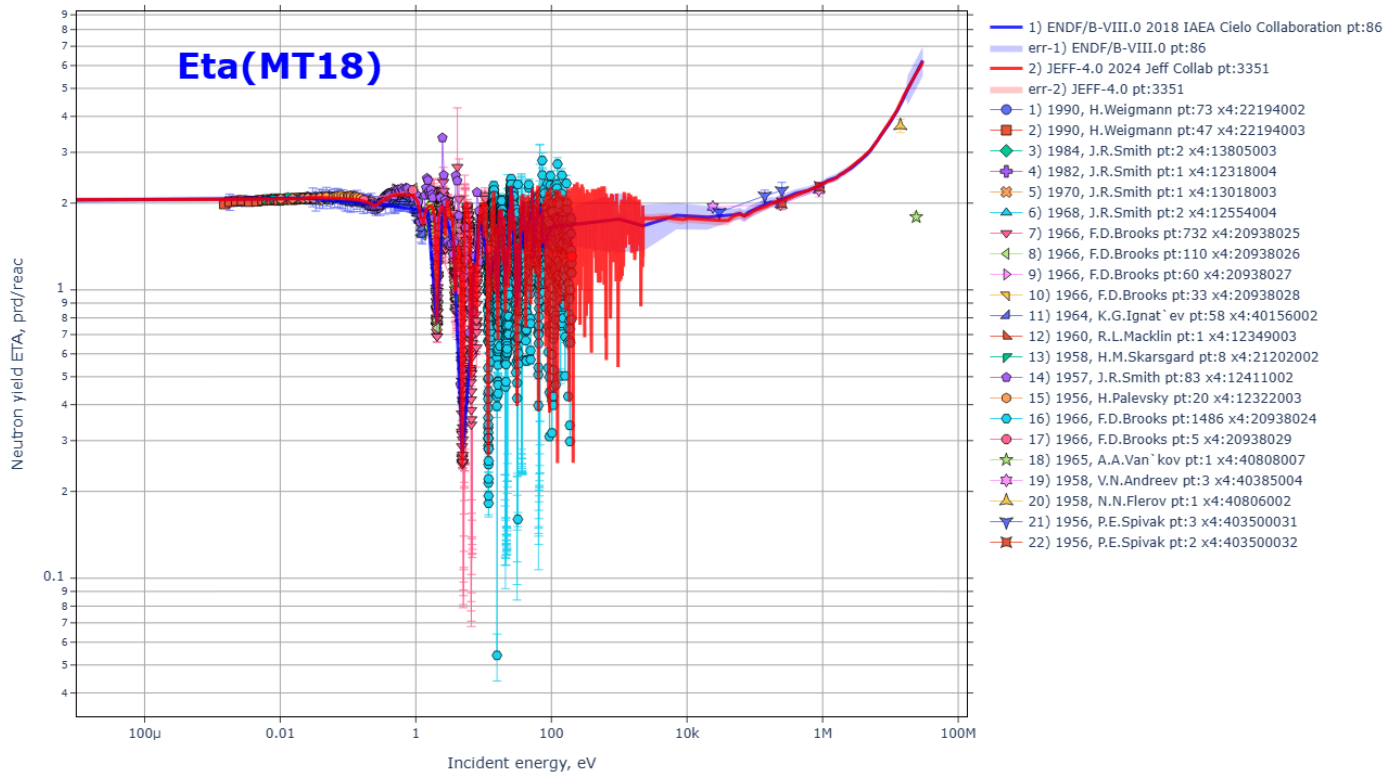
Reaction: 92-U-235(N,ABS),,ETA Quantity: NU:y=Data(En) Datasets: 22 Points: 2732
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:42:41



part2-A-eta/ EXFOR/ENDF Eta: neutron yield per nonelastic even

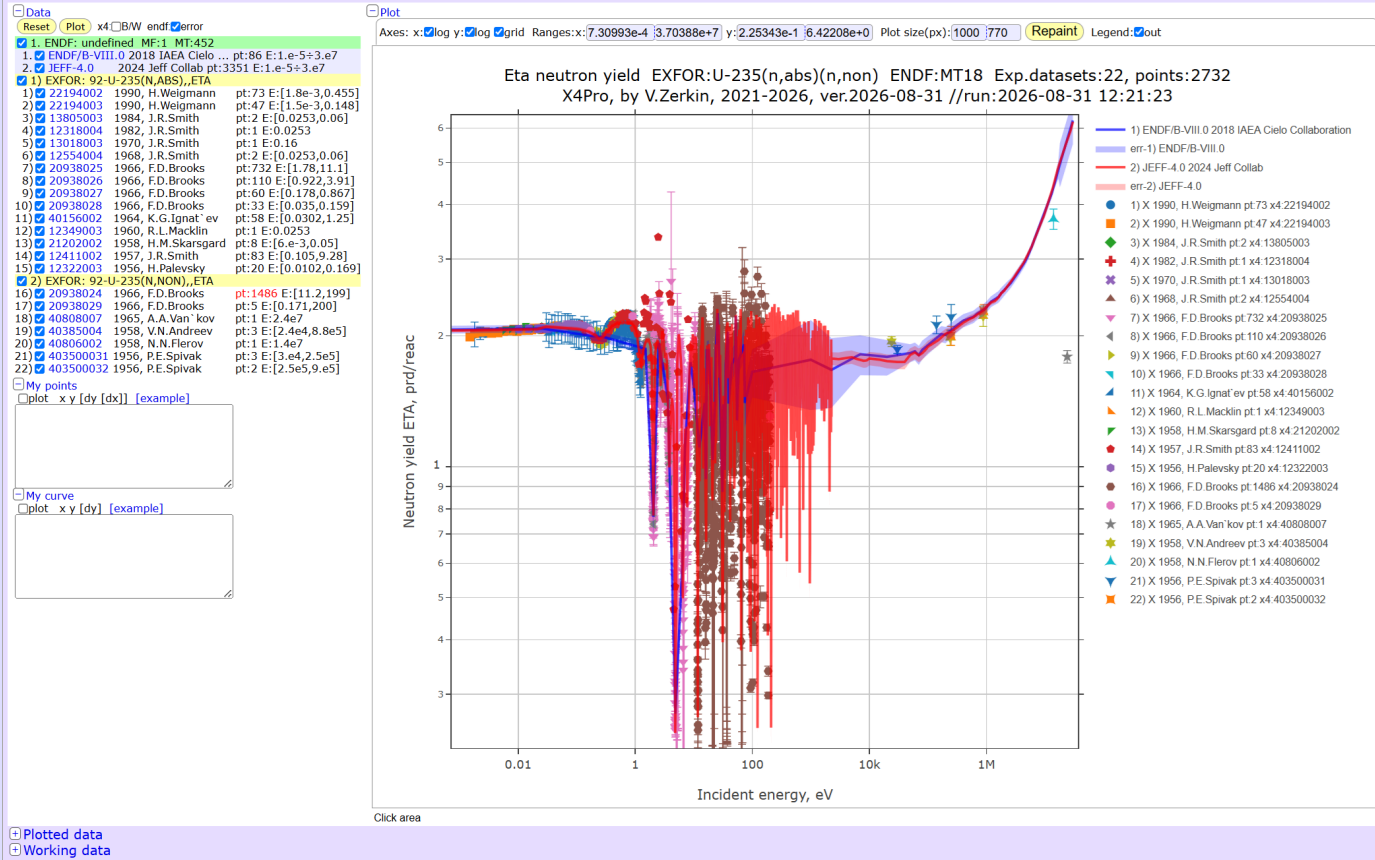
18) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF $\eta \pm \Delta\eta$

Eta neutron yield EXFOR:U-235(n,abs)(n,non) ENDF:MT18 Exp.datasets:22, points:2732
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-13 17:48:53



19) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33

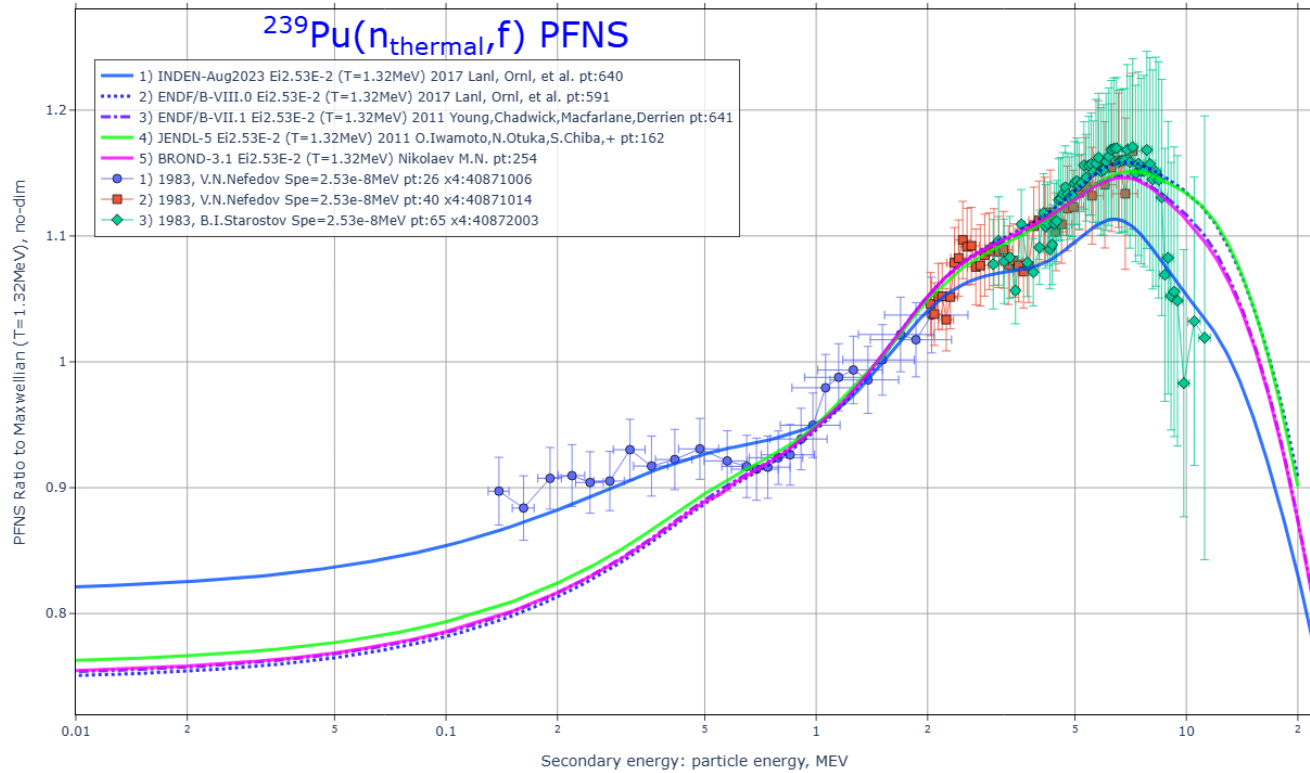
X4Pro - advanced plot
by V.Zerkin, 2025-2026



part2-B-pfns/ EXFOR/ENDF PFNS - energy spectrum of prompt fission neutrons

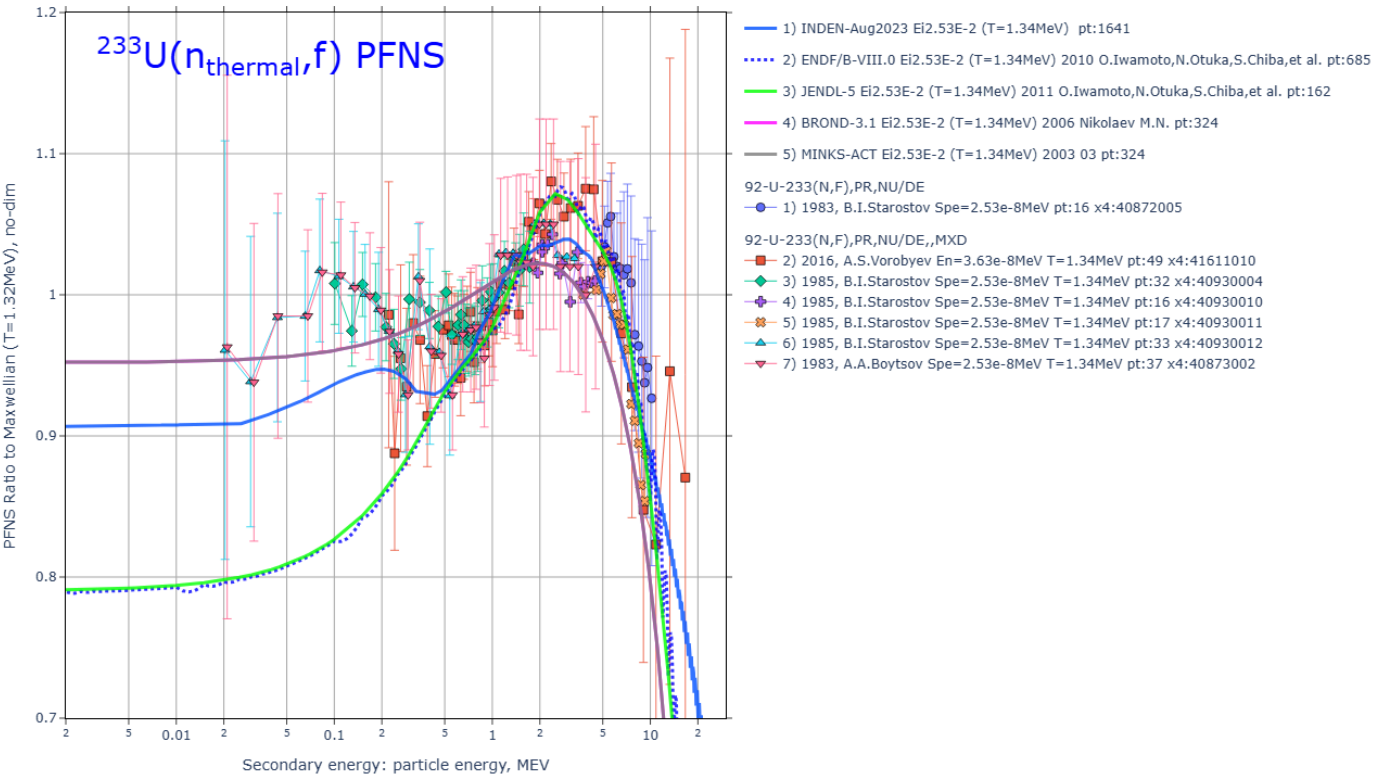
20) part2-B-pfns/out00/pfns239pu_t.html.png $^{239}\text{Pu}(n_{\text{thermal}},f)$ PFNS-ratio to Mxw.distr. (T=1.32MeV)

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:52:38



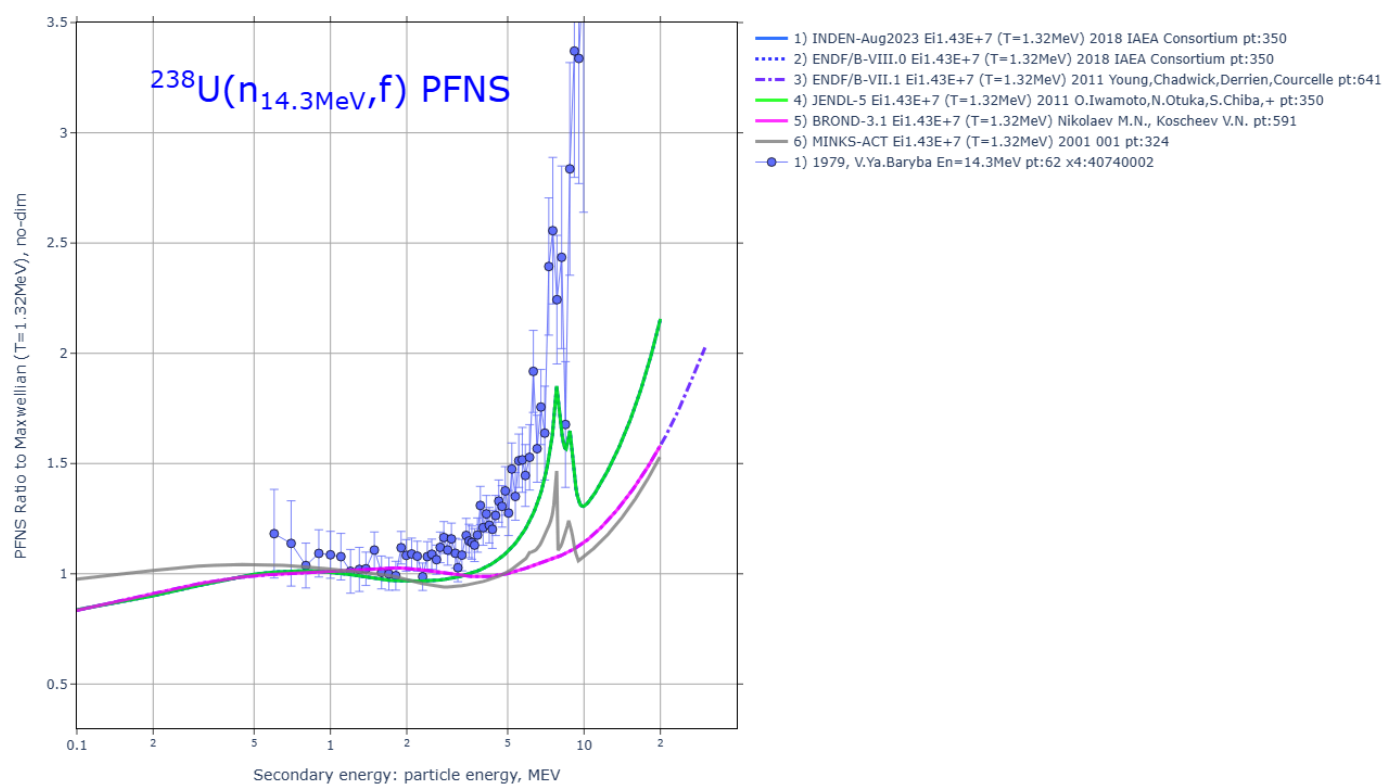
21) part2-B-pfns/out00/pfns233u_t.html.png ²³³U(n_{thermal},f) PFNS-ratio to Mxw. (T=1.34MeV)

Reaction:92-U-233(N,F),PR,NU/DE Quantity:MFQ:y=Data(Spe,E2) Datasets:7 Points:200
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:50:35



22) part2-B-pfns/out00/pfns238u14_3.html.png $^{238}\text{U}(n_{14.3\text{MeV}},f)$ PFNS-ratio to Mxw. (T=1.32MeV)

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 62
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:52:18



23) part2-B-pfns/out00/pfns239pu1_45.html.png ²³⁹Pu(n_{1.45MeV},f) PFNS-ratio to Mxw. (T=1.32MeV)

Reaction:94-PU-239(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:3/4 Points:136/165
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-13 17:53:01

